



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI PRANAV FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

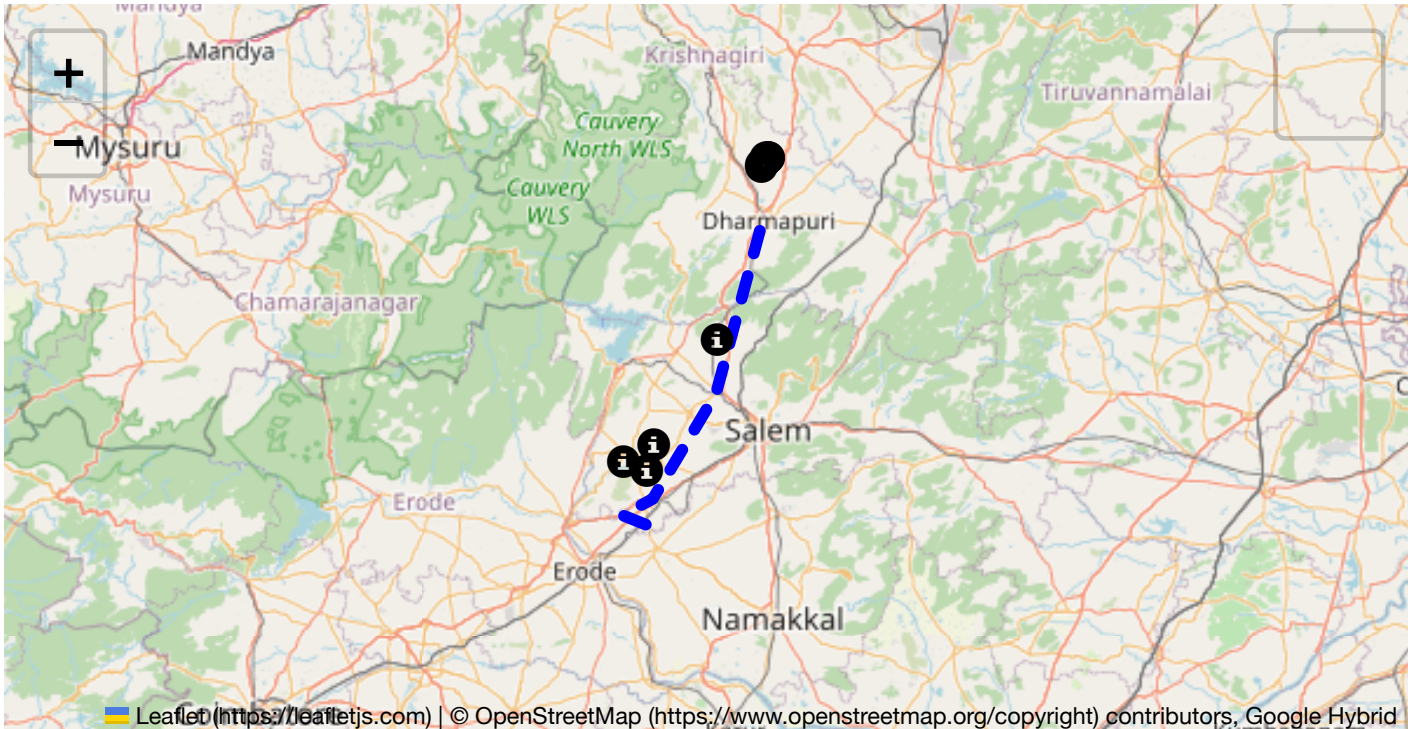
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

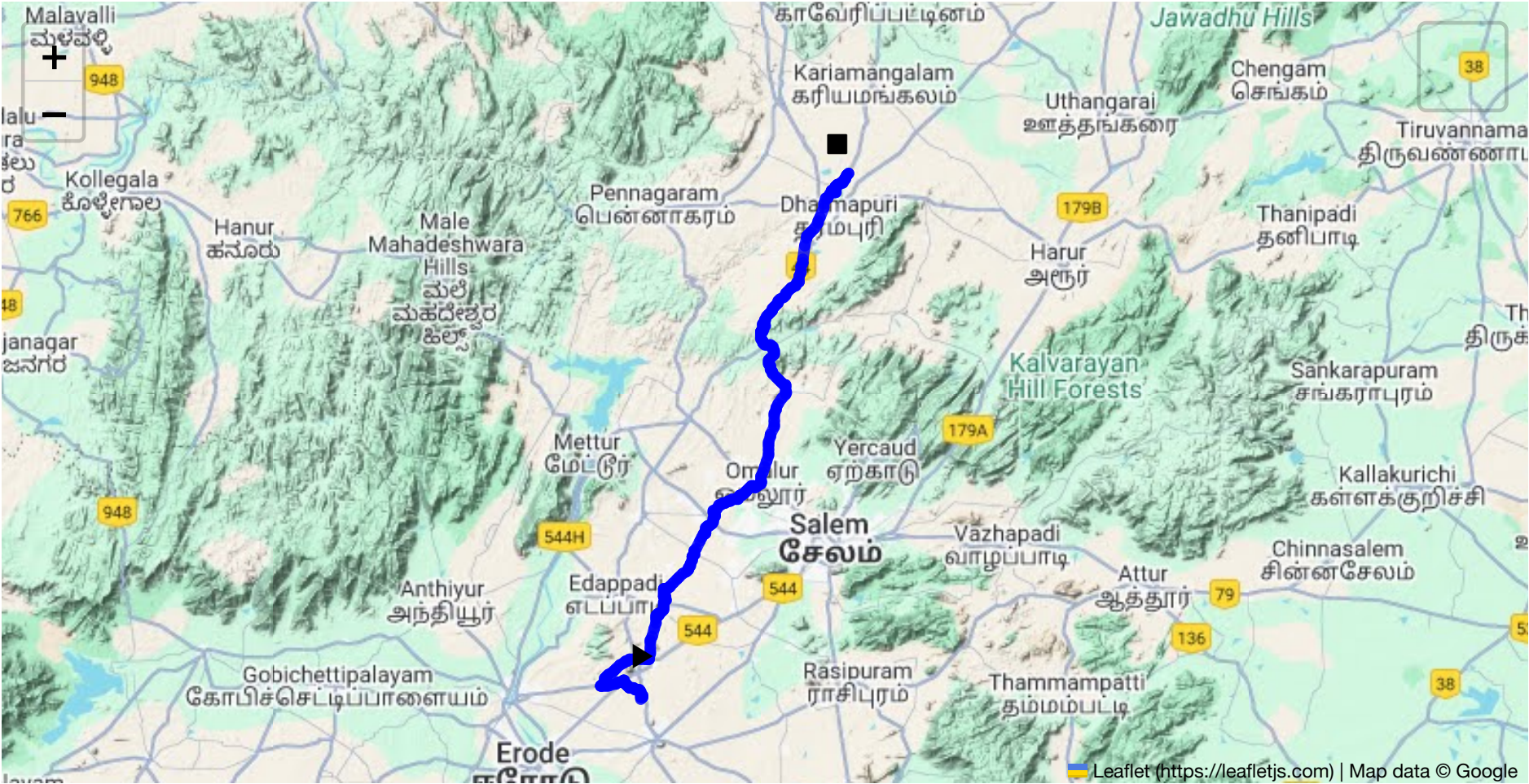
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 116.04 km
Estimated Duration: 2.4 hours
Adjusted Duration (Heavy Vehicle): 3.0 hours

Start: (11.4381, 77.8734)
End: (12.159297, 78.153654)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Kadagathur spans approximately 116.04 kilometers, passing through key areas like Bhavani Main Road, Salem, and NH 44 before reaching A.Reddihalli. This journey passes through both urban and rural landscapes, which may influence traffic and road conditions. The route primarily follows the Bhavani Main Road and SH 86, transitioning onto the National Highway 44, eventually leading to smaller state roads towards its destination.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences tropical weather, with hot summers (March to June), a monsoon season (July to October), and mild winters (November to February). The monsoon season can lead to heavy rainfall, potentially causing flooding and poor visibility. Drivers should prepare for slippery roads and possible roadblocks due to waterlogging, especially in low-lying areas.

3. Analysis of Traffic Patterns

Peak traffic hours are generally in the mornings (8:00 AM to 10:00 AM) and evenings (5:00 PM to 8:00 PM) as commuters travel to and from work. Areas like Salem, being a significant urban center, may experience more congestion. National Highway 44 is a busy route frequently used by heavy vehicles, which can lead to slow-moving traffic.

4. Assessment of Road Quality and Infrastructure

The roads on this route, particularly NH 44, are generally in good condition given their importance for interstate travel. However, some state highways and rural roads may have uneven surfaces, potholes, or narrow lanes which could pose challenges, especially after heavy rains.

5. Suggestions for Alternative Routes for Emergencies

In case of any major disruptions on NH 44, an alternative would be to use the nearby parallel state highways and rural roads, though they may be less equipped to handle heavy traffic. It's advised that drivers remain updated on real-time traffic and road conditions via GPS or local news.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transporting hazardous materials requires adherence to specific regulations, including securing proper permits and ensuring vehicles are equipped with necessary safety equipment. Compliance with hours of service for drivers and restrictions during certain times or routes may apply, especially in populated areas like Salem.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

While specific historical incident data should be sourced from local traffic authorities or safety boards, common issues might include accidents due to overloading or brake failures on winding roads. It is imperative to inspect vehicles thoroughly and to abide by safety measures.

8. Environmental Considerations and Sensitive Areas

This route crosses agricultural areas and small communities that could be vulnerable to spills or other environmental hazards. Protecting these areas involves careful adherence to safety protocols and immediate reporting of any incidents.

9. Analysis of Communication Coverage

While major towns and highways generally have good communication coverage, rural stretches might experience lower signal strength or dead zones. Ensuring communication devices are fully charged and having alternative contact methods, like satellite phones, could mitigate this risk.

10. Estimated Emergency Response Times

Emergency response times may vary, generally being quicker in urban areas like Salem but slower in rural regions. Coordination with local authorities for any pre-trip emergency planning is advised to ensure the fastest possible response times in the event of an incident.

12. Overall Summary of Risk Assessment

This route is reasonably safe but not without its challenges. Weather conditions during the monsoon, rural road quality, and high traffic on NH 44 are primary concerns. Ensuring vehicle safety, familiarizing oneself with alternative routes, and maintaining open lines of communication are critical in managing and mitigating potential risks effectively. Regular training for drivers on safety protocols and emergency response is also strongly recommended to minimize risks associated with transporting hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	U-Turn	High	12.1790414, 78.1722626	10 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	High	11.44900, 77.87406	15 KM/Hr
4	Turn	High	11.45353, 77.85697	15 KM/Hr
5	Turn	Medium	11.45853, 77.85139	30 KM/Hr
6	Turn	Medium	11.45906, 77.85132	30 KM/Hr
7	Turn	High	11.46307, 77.84941	15 KM/Hr
8	Turn	High	11.45542, 77.81725	15 KM/Hr
9	Turn	High	11.45636, 77.81678	15 KM/Hr
10	Turn	Medium	11.49383, 77.88501	30 KM/Hr
11	Blind Spot	Blind Spot	11.49400, 77.88538	10 KM/Hr
12	Turn	High	11.49469, 77.88488	15 KM/Hr
13	Turn	High	11.55618, 77.89789	15 KM/Hr
14	Turn	Medium	11.55629, 77.89888	30 KM/Hr
15	Turn	Medium	11.58228, 77.90841	30 KM/Hr
16	Turn	Medium	11.58290, 77.90783	30 KM/Hr
17	Turn	High	11.58340, 77.90637	15 KM/Hr
18	Turn	Medium	11.63832, 77.94818	30 KM/Hr
19	Turn	Medium	11.64161, 77.95177	30 KM/Hr
20	Turn	Medium	11.64226, 77.95200	30 KM/Hr
21	Turn	Medium	11.64716, 77.95118	30 KM/Hr
22	Turn	High	11.67868, 77.96836	15 KM/Hr
23	Turn	Medium	11.70077, 77.97957	30 KM/Hr
24	Turn	High	11.70223, 77.97860	15 KM/Hr
25	Turn	Medium	11.71402, 77.99586	30 KM/Hr
26	Turn	Medium	11.71918, 78.01127	30 KM/Hr
27	Turn	Medium	11.72565, 78.01770	30 KM/Hr
28	Turn	Medium	11.73769, 78.03451	30 KM/Hr
29	Turn	Medium	11.73785, 78.03551	30 KM/Hr
30	Turn	High	11.73651, 78.03871	15 KM/Hr
31	Turn	Medium	11.74271, 78.04717	30 KM/Hr
32	Turn	Medium	11.78017, 78.05790	30 KM/Hr
33	Turn	Medium	11.86953, 78.08386	30 KM/Hr
34	Turn	Medium	11.88019, 78.08158	30 KM/Hr
35	Turn	Medium	11.91282, 78.05854	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
36	Turn	Medium	11.93757, 78.05319	30 KM/Hr
37	Turn	Medium	11.95816, 78.04976	30 KM/Hr
38	Turn	Medium	11.95870, 78.04990	30 KM/Hr
39	Turn	High	11.96031, 78.04945	15 KM/Hr
40	Turn	Medium	11.96291, 78.05131	30 KM/Hr
41	Turn	Medium	11.96390, 78.05138	30 KM/Hr
42	Turn	Medium	11.96453, 78.05092	30 KM/Hr
43	Turn	High	11.96562, 78.04928	15 KM/Hr
44	Turn	High	11.96667, 78.04934	15 KM/Hr
45	Turn	Medium	11.96775, 78.05056	30 KM/Hr
46	Blind Spot	Blind Spot	12.15370, 78.14994	10 KM/Hr
47	Blind Spot	Blind Spot	12.15378, 78.14971	10 KM/Hr
48	Turn	Medium	12.17027, 78.16812	30 KM/Hr
49	Blind Spot	Blind Spot	12.17897, 78.17236	10 KM/Hr
50	Blind Spot	Blind Spot	12.15359, 78.15011	10 KM/Hr
51	Turn	Medium	12.15347, 78.15030	30 KM/Hr
52	Turn	Medium	12.15265, 78.15265	30 KM/Hr
53	Turn	High	12.15264, 78.15322	15 KM/Hr
54	Blind Spot	Blind Spot	12.15889, 78.15504	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
7	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
8	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
10	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
11	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
12	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
13	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
14	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
15	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
16	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
17	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
18	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
19	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
20	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
21	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
25	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
26	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
27	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
9	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
22	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
24	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
28	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 12.1790414, 78.1722626



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45853, 77.85139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45906, 77.85132



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45636, 77.81678



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49383, 77.88501



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49400, 77.88538



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.55618, 77.89789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58340, 77.90637



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63832, 77.94818



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64161, 77.95177



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64226, 77.95200



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64716, 77.95118



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67868, 77.96836



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70077, 77.97957



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70223, 77.97860



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71402, 77.99586



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71918, 78.01127



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72565, 78.01770



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73769, 78.03451



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73785, 78.03551



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73651, 78.03871



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74271, 78.04717



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.78017, 78.05790



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86953, 78.08386



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88019, 78.08158



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91282, 78.05854



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93757, 78.05319



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95816, 78.04976



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95870, 78.04990



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96031, 78.04945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96291, 78.05131



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96390, 78.05138



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96453, 78.05092



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96562, 78.04928



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96667, 78.04934



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96775, 78.05056



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.15370, 78.14994



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.15378, 78.14971



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.17027, 78.16812



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.17897, 78.17236



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.15359, 78.15011



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.15347, 78.15030



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.15265, 78.15265



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.15264, 78.15322



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.15889, 78.15504

